

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3				5-6 marks A comprehensive account of all 3 areas is provided. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks A comprehensive account is given for 2 out of the 3 areas i.e. experimental set up, data collection and analysis or a limited account of all 3 areas is provided. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks A comprehensive account is given for 1 out of the 3 areas i.e. experimental set up, data collection and analysis or a limited account of 1 or 2 areas is provided. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)	I	Substitution: $E = hf = h \frac{c}{\lambda}$ $E = (6.63 \times 10^{-34}) \left(\frac{3 \times 10^8}{1 \times 10^{-12}} \right) (1)$ $= 2.0 \times 10^{-13} \text{ [J]} (1) (\text{or } 1.24 \text{ MeV})$	1	1		2	2	
			II	$E = (6.63 \times 10^{-34})(1.8 \times 10^9) = 1.2 \times 10^{-24} \text{ [J]}$		1		1	1	